Resource management and scale transfer: the contribution of multi-agent systems

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1 - Introduction

The management of natural and renewable resources must take account of both resource dynamics and social dynamics. However, in order to understand these dynamics, measurements must be made which raise problems of scale transfer : how can local processes be scaled up to produce a more global description applicable to a whole region ? Rather than reasoning in terms of scale transfer, this article examines the relations between different levels of organization. Dynamics are described at the level of the plot, the farmer, the village, the forest, the irrigated zone, the social group, the region, etc... The problem is to understand how the dynamics identified at each different level link up with each other.

This article presents a modelling method: the multi-agent system (MAS) which provides a response to this problem. First of all, multi-agent systems are presented and the notion of level of organization is discussed. We then present an example concerning the spatial dynamics of change in a rural area. We propose two alternative models to simulate the same spatial dynamics. The two models are based on processes represented at various levels.

2. Multi-agent systems and levels of organization

2.1 Multi-agent systems

Multi-agent systems offer a modelling method based on the principles of distribution and interaction (Bond et Gasser 88, Lesser 95, O'Hare and Jennings 96, Ferber 95). To model a complex phenomenon, different interacting entities with specific behaviour patterns are represented. The computer entities (agents) perceive their environment and are able to act upon it, are able to communicate by sending messages and to make representations of the world. The principles correspond to a bottom-up approach for the representation of reality. Knowledge is represented at the microscopic level and phenomena are observed at the macroscopic level. We speak of an emerging phenomenon. For several years now, multi-agent systems have been used to study a circular relation between levels. The macroscopic level is represented in the form of a group or macro-agent. The question is to determine how the interactions between entities at the microscopic level can bring about phenomena at the macroscopic level and, conversely, how the macroscopic level affects dynamics at the microscopic level (Rouchier et al., 1998). Modern computer platforms are now able to take account of these different levels (Ferber and Gutknecht 1998, Marcenac et al. 1998). Lastly, it is possible to represent interactions between agents at different levels in a single multi-agents system. For example, a Farmer agent interacts with a Forest agent while a Village agent interacts with a Tree agent. Here, we move on from the concept of simple hierarchies to that of dynamic interlinking hierarchies which corresponds to the reality of natural and renewable resource management.

Widely used in telecommunications and distributed systems, multi-agent systems are also used in the fields of resource management and the environment, (Bousquet 94), (Le Page and Cury

1996), {Barreteau and Bousquet 98}. Some researchers are focusing on the simulation of biophysical processes, others on social dynamics. In both cases, multi-agent systems are often used to represent spatial dynamics. In the future, these methods will find numerous applications in fields such as landscape ecology, for which distributed methods with cellular automata are already used (Gaylord and Nishidate 96). Multi-agent systems offer greater flexibility and broader potential than cellular automata.

2.2. Cormas (Common-pool resources and multi-agent systems)

To compare the two approaches for simulation of spatial dynamics by MAS, we worked under CORMAS (Bousquet at al., 1998). This simulation environment comprises an organization of Smalltalk language classes and a series of interfaces for programming multi-agent systems for simulation. CORMAS includes various sets of programs. The first is used to model agents and interactions; these interactions are mediated either via a simulation space (i.e. a set of cells), or via messages. The second set of programs is used to control simulation dynamics while the third serves to define the various viewpoints of the observer.

Three types of agent can be defined in the CORMAS environment. The first is called a situated agent. A situated entity has spatial references (the cell). It is characterized by its field of perception. Departure and arrival methods enable the user to program the agent's departure from the cell and its arrival at another cell. The second type of agent is called a communicating agent. A communicating agent possesses a mailbox and basic programs to receive and send messages. The third type of agent is the situated and communicating agent, which possesses the combined characteristics of the first two. To create an agent, the user applies the notion of inheritance specific to the object-oriented concept. The agent created by the user will be a specialization either of the SituatedObject class, the CommunicatingObject class or the SituatedCommunicatingObject class.

These agents can interact in two ways. The first concerns the way that space is taken into account. This space comprises a grid of cells, as is the case with cellular automata, corresponding to a space upon which the interactions between agents take place. The cell thus possesses a list of situated agents positioned within the cell. It provides primitives for access to neighbouring cells located at variable distances. The second mode of interaction is modelled by messages that the agents send to each other via a transmission channel. All communicating agents are linked by a channel which serves to transmit messages.

The CORMAS environment is equipped with a range of tools to program the viewpoint from which the system is observed. Three types of interface can be used: (1) graphs; (2) a grid to view all cells in the space and the situated agents positioned inside them and which represents the geographical area under analysis; (3) a link observer, which can be used to view message exchanges between agents by drawing dynamic graphs.

3. Dynamics on different spatial scales

In order to use MAS's to simulate spatial dynamics we must first determine what elements should be considered as spatial agents and what the rules of their spatial interaction should be. The solution generally chosen is to consider “physical” agents and to simulate their interactions. This approach to spatial dynamics corresponds closely to that of the MAS simulation tool, with behaviours allocated to situated agents. We thought it would be interesting to move beyond these representation and to simulate spatial dynamics on the basis of geographical entities (forest, town, road, etc.), endowed with a specific behaviour[1]. In order to evaluate the methodological gain of this new approach, we compared the two approaches for simulation of spatial dynamics with MAS's, (1) by agents situated in the cell and (2) by geographical entities (Bousquet et Gautier, 1998), by applying them to the same example. After presenting the theoretical example of spatial dynamics, and then the simulation environment, we will examine the methodological contribution of this comparison in terms of relations between organization levels.

3.1 The theme

The two approaches for simulation of spatial dynamics with MAS's are compared for a same state of a same territory, situated in a hypothetical savannah. This state of the territory is subject to the same dynamic processes, of which two versions exist, depending on whether the dynamics are observed from the viewpoint of the actor in the cell or from that of the spatial entity. The state of the rural territory presents the classic trilogy of rural development: ager, saltus, sylva. It is managed by farmer-herders who set up villages in fertile zones, clear fields around the village up to an expansion limit governed by the distance of dwellings from the fields or by the size of the fertile zone. Once this limit has been reached, a new village is set up in a new fertile zone. In parallel with this process of agricultural development, herds of livestock graze on the savannah around the fields. Their size grows in proportion to that of the villages. This increase in herd size places growing pressure on the pasture zones, resulting in the clearing of forested areas and hence new opportunities for creating additional villages.

3.2 Approaches

In methodological terms, space is considered as a resource zone, comprising a grid of cells in which the actors move and act. The cells are characterized by two attributes (variables): state and fertility. There are four state values (forest, savannah, field, village) (Figure 1) and two fertility values (fertile, infertile).
The processes which generate the spatial dynamics of this territory can be seen from two viewpoints.

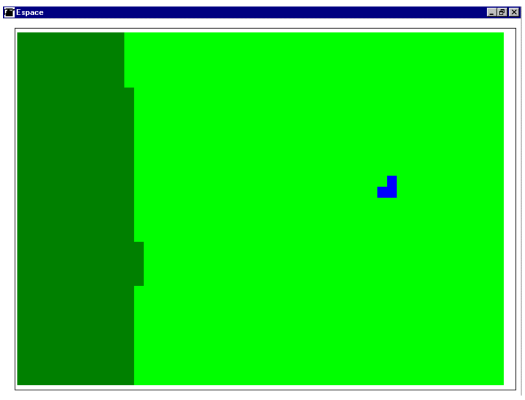

Figure 1 The initial state of territory. Dark green (left side of the picture) represents the forest, light green the savannah and blue (three cells) the village

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[1] Work performed with the Montpellier-based "SMA Spatialisés" network, created on the initiative of S. Lardon and R. Lifran (INRA Montpellier)

3.2.1 The viewpoint of the actors managing the natural resources

Each agent represents a family of farmer-herders. The agent settles in a fertile zone, builds a house, clears the land around the house to create as many fields as are required to produce food for the family, and takes a family herd to graze on the savannah. An increase in the population of farmer-herders leads to the construction of new houses adjacent to the initial house, the clearing of fields around the new village and an increase in the size of herds in proportion to the needs of the population. Village expansion stops when the last farmer-herders to arrive cannot find any fertile fields around the village or when the distance between the fields and the village becomes to large. At this point, a new fertile zone is then colonized according to the same rules of behaviour, though the increasing herd size leads to a certain amount of forest clearance (Figure 2).

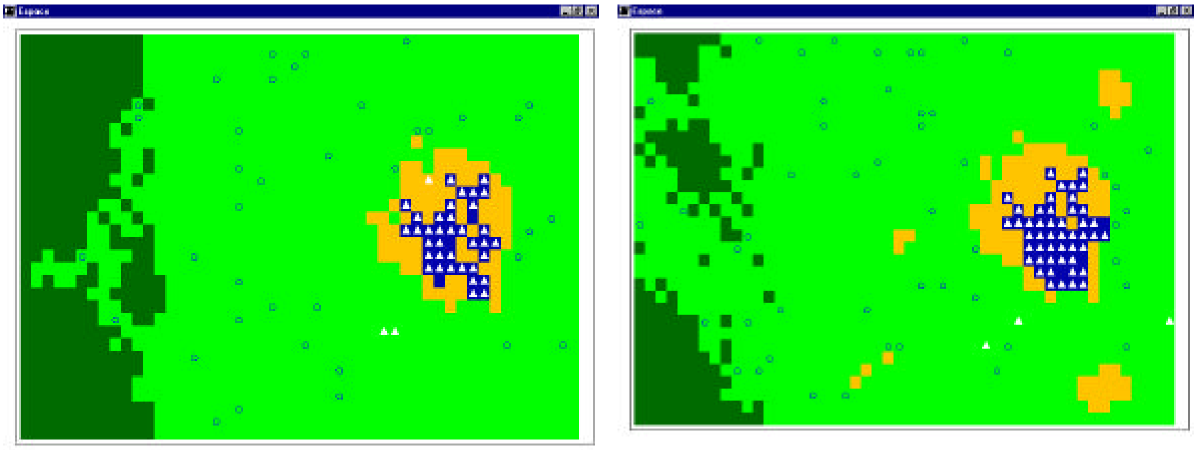

Figure 2. Snapshots at time step 50 and 150. The white triangles represent the Farmer agents, the black circles represent the Herd agents. The yellow color represent the fields.

Two classes of agents have been created. The Farmer agent: it inherits from the SituatedObject class. It has a field attribute which is a collection of the fields it cultivates. It behaves as follows: at each time step, the agent, which has a perception radius of three cells, looks for a cell which is both in the "savannah" state and "fertile". If it finds a cell of this kind, it transforms the cell into a "field" and adds it to its field collection. When an agent has found three fields, it installs a member of its family close to its dwelling. In order to do so, a "field" cell is transformed into a "village" cell.

The Herd agent: it inherits from the SituatedObject class. At each time step, the agent, which has a perception radius of one cell, looks for a cell which is neither a "village" cell nor a "field" cell. It moves randomly among the cells corresponding to this condition. However, when it arrives in a "forest" cell, it transforms it into a "savannah" cell.

3.2.2. A spatial viewpoint

The agents are no longer farmer-herders or herds, but geographical entities (village, field, savannah, forest) endowed with specific behaviours. In the spatial dynamics approach with geographical entities, we focus on the entities composing the territory, village, field, savannah and forest (elements) and on the relations between them (interactions) resulting from biophysical processes or development by man (Figure 3).

The cells of the space are the same as for the previous simulation. Four classes of agents are created: Village, Field, Forest and Savannah. These four classes of agents have points in common. They inherit from the CommunicatingObject class and they possess a "components" attribute which represents the collection of cells in the resource zone that the agent controls. Each agent is able to consult its mailbox and to appropriate the cells sent to it. A Message class is created. It possesses a "data" attribute for exchanging cells between agents in accordance with the stated rules.

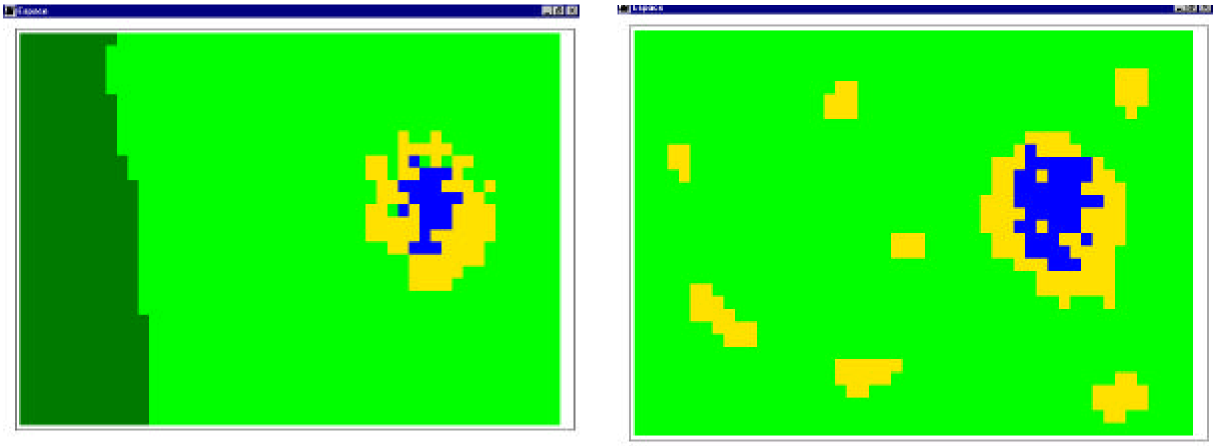

Figure 3. Two snapshots at time step 8 and 15.

The Village agent can calculate a need for farming land according to the size of the village. The village size is equal to the quantity of cells it comprises. At each time step, the Village agent consults it mailbox and adds any cells that it has received to its "components" collection. It sends a message to the Field agent to request cells and to the Savannah agent to ask for three times more cells (i.e., three field cells for one village cell).
The Field agent consults its mailbox and adds any cells that it has received to its "components" collection. If it receives a message from the Village agent, it transfers the requested cells, provided that it controls four times the number of cells requested.
The Savannah agent consults its mailbox and adds any cells that it has received to its "components" collection. If it receives a message from the Village agent, it transfers the requested cells and sends a message to the Field agent. To this end, the Savannah agent counts the number of cells it comprises which are adjacent to a "field" or "village" cell in a "fertile" fertility state. If the number obtained is not sufficient to satisfy demand, the Savannah agent looks for cells which satisfy the fertility criterion only, no longer taking account of their position. It then sends a message to the Forest agent asking for half the quantity of cells that it has just given away.
The Forest agent consults its mailbox and adds any cells that it has received to its "components" collection. To this end, the Forest agent looks for cells adjacent to savannah cells among the cells it comprises.

3.3 Discussion

3.3.1. Approaches to space

By establishing relations between spatial processes and the behaviours of actors via the same combinations of biophysical and human processes, it is possible to compare these two

viewpoints, though more from a methodological angle than with respect to the results obtained.
When examining spatial dynamics from the viewpoint of the actors, we focus on individuals acting on space, farmer-herders or herds (elements), and on the relations between these actors and the space and its resources (interactions). Spatial organization results from the impact of the action of the actors on the space, with the space affecting the behaviour of these actors via constraints or advantages. In simulation, space is integrated at the cellular level.

When spatial dynamics are examined from the viewpoint of spatial entities, space is perceived as the product of interactions between the spatial entities, these being linked to the behaviour of the actors. In this approach, space is integrated at the overall level of the geographical entity and no longer at the cellular level in terms of interactions between agents acting upon the space which surrounds them.

3.3.2. Levels of organization

The example presents two alternatives. The first model simulates interactions at the microscopic level: the entities are Herd and Farmer agents, the space is broken down into cells. This corresponds to a simple bottom-up approach. The second model represents the behaviour of spatial aggregates: the Forest, Savannah, Field and Village agents. Each of these agents is taken to represent an overall dynamic which controls all spatial cells. This level of integration makes it possible to endow each geographical entity with characteristics which affect spatial dynamics (such as the form of the entity or its position in a territorial pattern) and to introduce these characteristics into the rules of interaction (e.g., the more compact and isolated it is, the more easily a patch of forest will give up cells to the savannah). Hence, an aggregate is more than the sum of its parts. A forest entity is not only the collection of "forest" cells. There may be specific dynamics due to overall properties (form for example). Similarly, the behaviour of the village entity is not simply the sum of the behaviours of Farmer agents. For example, the Village agent is capable of calculating an indicator corresponding to collective satisfaction and of asking for savannah cells only when collective satisfaction is low.
This didactic exercise shows us that dynamics are expressed at every level. In other simulations, we could make the various levels of organization coexist side by side. For example, we could do simulations with Farmer, Village, Cell and Savannah agents. The Farmer agents ask for space from the Savannah agent which then adjusts its contours to take account of the space taken from it and modifies the cells it controls. For social dynamics, we can consider that the Village agent is responsible for calculating the surface area of fields per inhabitant to provide an indicator of collective satisfaction. If the Village is not satisfied, the Farmer agents go out to look for new fields.
For Allen and Hoekstra (1992) “there is plenty of room for entities from almost any type of ecological system to be contained within an entity belonging to any other class of systems”. Therefore they build “a layer cake” of ecological system. For them the most interesting questions will involve slicing the cake diagonally. Diagonal slices allow to see how patterns defined at a given scale (the rural area at landscape level) are influenced by dynamics described in terms of organism (a farmer) and by dynamics described in terms of ecosystems (a forest).

4. Conclusion

In this article, we have presented a method to model and simulate natural and renewable resource management. We have shown how it is possible to represent knowledge on several levels of organization. Rather than raise the question of scale transfer, these methods enable us to focus on the relations between dynamics on several levels. We are able, for example, to model the spatial dynamics associated with the actions of farmers on their land, integrating the interactions between farmers and the behaviour of upper organization levels such as the village. From a spatial point of view one can represent dynamics at several levels from the plot to the region by modelling different spatial entities.

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